

CONSTRUCTION D'UNE VILLA A DJERBA HOUMT SOUK

B					
A					
0	03.11.2025	1ère diffusion	H.A	H.A	H.A
IND.	DATE	MODIFICATIONS	ETABLI PAR	VERIFIE PAR	APPROUVE PAR

CARNET ARMATURES POUTRES ET RAIDISSEURS
PH-RDC

N° DE PLAN

ARM 01

0

INDICE

MAITRE D'OUVRAGE
ARCHITECTE
MAITRE D'OEUVRE RENOVATION
ENTREPRISE GROS OEUVRE
BET STRUCTURE
BUREAU DE CONTROLE



HA CONCEPT

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Mail : engineer.haithem.ayadi@gmail.com

REF. AFFAIRE :	FORMAT :	PHASE :
25040	A1	EXE
DATE 1ère DIFFUSION :	ZONE OU BLOC :	ECHELLE :
03 11 2025	G	$\frac{1}{50}$ & $\frac{1}{25}$

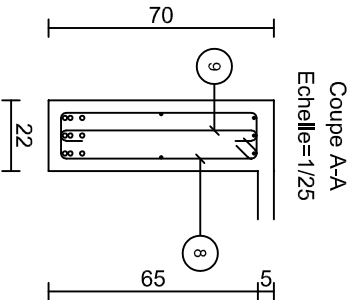
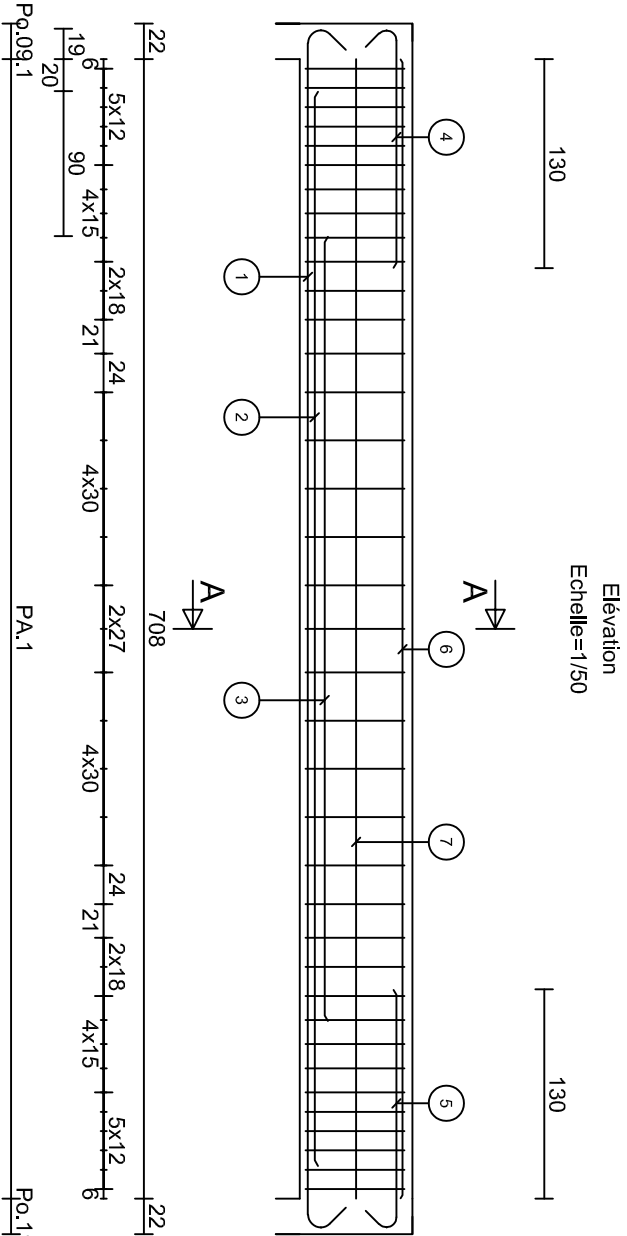
Béton
C25/30

Aciers non
soudés

VILLA CHAHIR	PA.1	Béton=1,12 m3 Acier=107,7 kg d=96,5 kg/m3 Fi=9,5 mm CoF=1,1 m² Densité acier: 7850,00 kg / m3	Eb=3,5 cm Ei=5,0 cm Ei=3,5 cm	1 — 1
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Section : 22 x 70ht

fck= 25 MPa fyk= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



Barre	Lg	Forme
1 3HA14	793	
2 3HA14	668	
3 3HA14	488	
4 3HA12	168	
5 3HA12	168	
6 3HA8	708	
7 2HA8	708	
8 37HA6	162	
9 37HA6	78	
Barre	Lg/Poids	
HA6	88,8/19,7	
HA8	21,2/8,4	
HA12	10,1/9,0	
HA14	58,5/70,7	

Béton
C25/30

Aciers non
soudés

VILLA CHAHIR	PA.2	Béton=1.24 m3 Acier=195.3 kg d=157.4 kg/m3 Ft=10.0 mm Cof=12.2 m² Densité acier: 7850.00 kg / m3	Ed=3.5 cm Et=5.0 cm Et=3.5 cm	1
PH RDC				

Section : 22 x 80ht

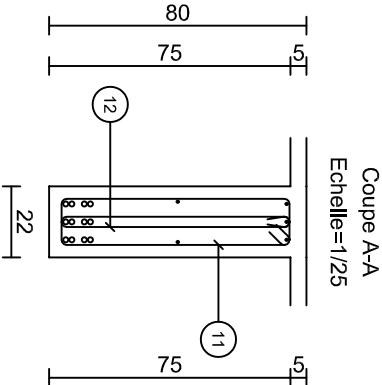
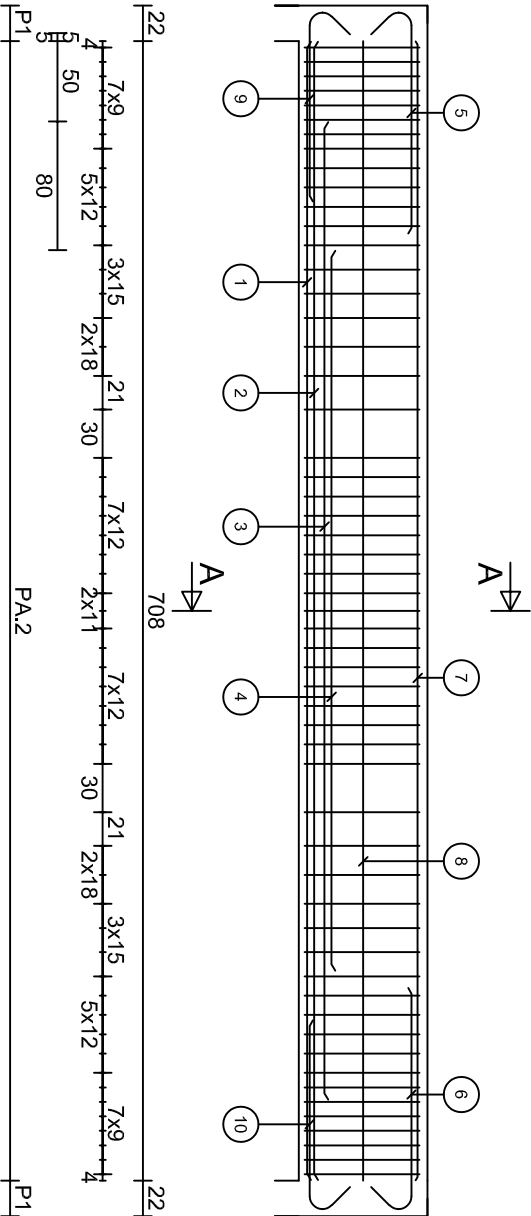
fck= 25 MPa ftk= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1

Elevation

Echelle=1/50

120

120



Barre		Lg	Forme
1	3HA16	708	708
2	3HA16	708	708
3	3HA16	608	608
4	3HA16	448	448
5	3HA14	164	164 139 16.2 135°
6	3HA14	164	164 135° 139 16.2 135°
7	3HA8	708	708
8	2HA8	708	708
9	3HA14	144	144 135° 16.2 118 135°
10	3HA14	145	145 119 16.2 135°
11	55HA6	182	182 15 16.2 135°
12	55HA6	161	161 71
Barre		Lg/Poids	
HA6	HA8	188.8/41.9	35.4/14.0
HA8	HA14	18.5/22.4	74.2/117.0
HA14	HA16		

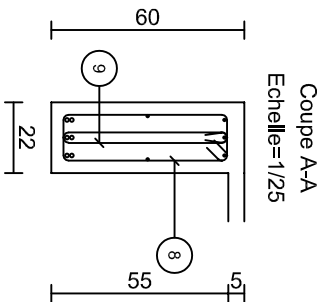
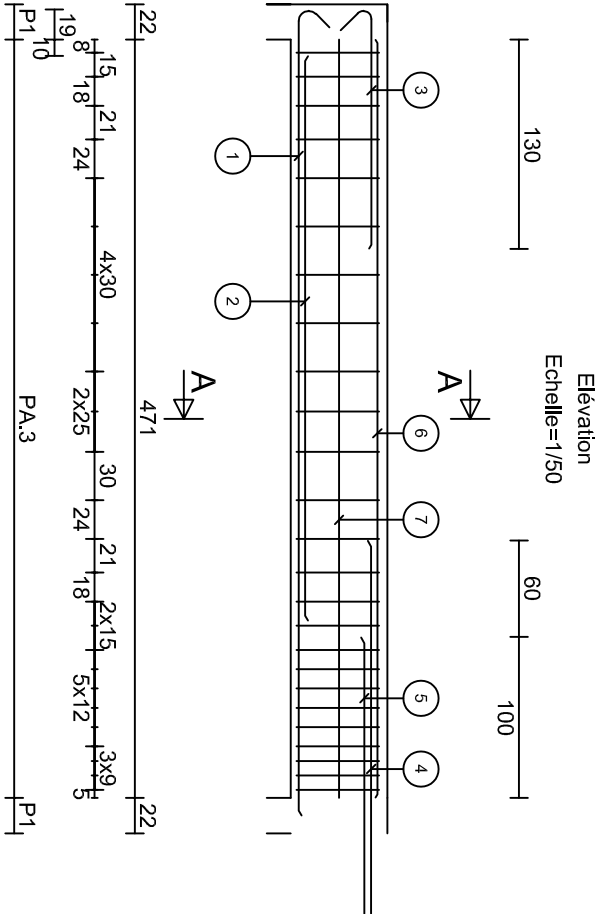
Béton
C25/30

Aciers non
soudés

VILLA CHAHIR	PA.3	Béton=0.64 m3 Acier=67.7 kg d=106.2 kg/m3 F _i =8.9 mm Cof=6.5 m² Densité acier: 7850.00 kg / m3	Ed=3.5 cm Ei=5.0 cm Ei=3.5 cm	1 5
PH RDC				

Section : 22 x 60ht

fck= 25 MPa f_{yk}= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



Barre	Lg	Forme
1 3HA12	521	135° 13 501
2 3HA12	351	351
3 3HA10	170	148 10 135°
4 3HA14	362	362
5 3HA14	222	222
6 3HA8	471	471
7 2HA8	471	471
8 25HA6	142	15 15
9 25HA6	121	51

Barre	Lg/Poids
HA6	65.8/14.6
HA8	14.1/5.6
HA10	5.1/3.1
HA12	26.1/23.2
HA14	17.5/21.2

Béton
C25/30

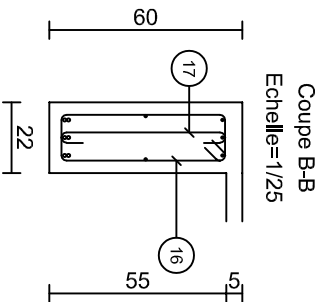
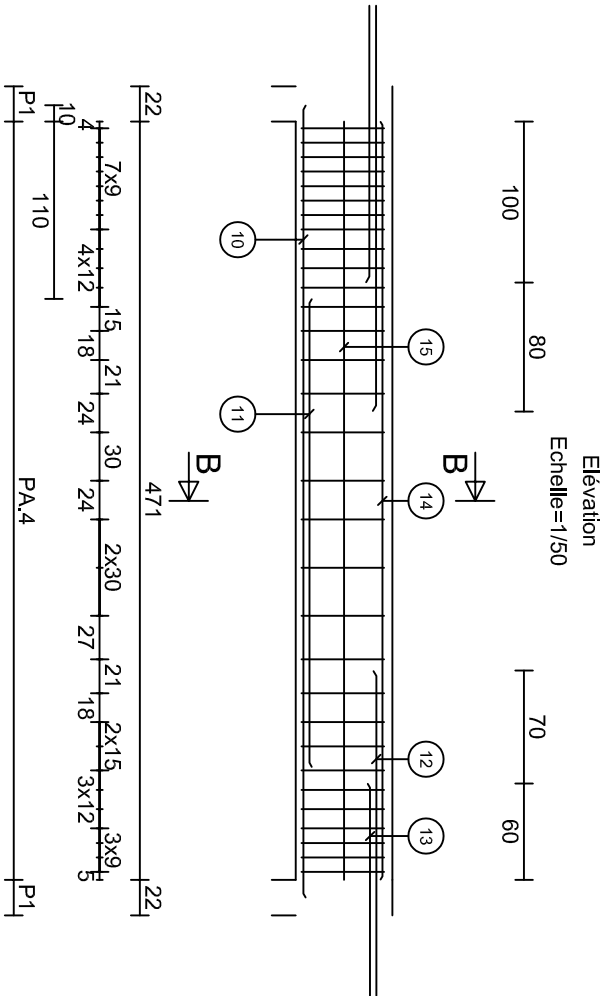
Aciers non
soudés

VILLA CHAHIR	PA.4	Béton=0.62 m3 Acier=45.7 kg d=73.3 kg/m3 F _i =8.7 mm Cof=6.5 m² Densité acier: 7850.00 kg / m3	EB=3.5 cm EH=5.0 cm EI=3.5 cm	2 5
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Section : 22 x 60ht

fck= 25 MPa fyk= 400 MPa Classe de ductilité A Coupe feu R 60

Classe d'exposition: XC1



Barre	Lg	Forme
10	492	492
11	291	291
12	322	322
13	142	142
14	471	471
15	471	471
16	142	15 15
17	68	51

Barre	Lg/Poids
HA6	65.1/14.5
HA8	14.1/5.6
HA10	27.8/17.1
HA12	9.7/8.6

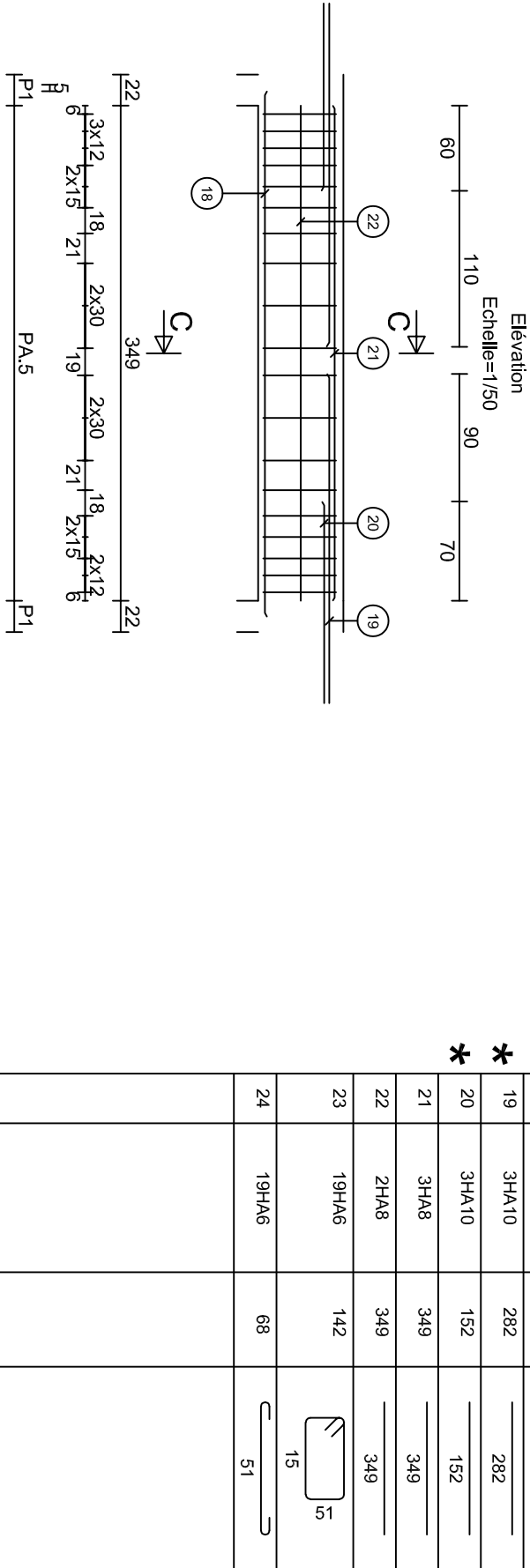
Béton
C25/30

✱ Aciers non
soudés

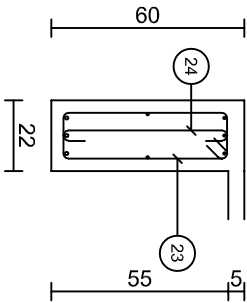
VILLA CHAHIR PH RDC	PA.5	Béton=0.47 m3 Acier=27.9 kg d=59.4 kg/m3 F _t =8.1 mm Cof=4.8 m² Densité acier: 7850.00 kg / m3	EB=3.5 cm EI=5.0 cm EI=3.5 cm	3 5
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Section : 22 x 60ht

fck= 25 MPa f_{yk}= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



Coupe C-C
Echelle=1/25



Barre		Lg	Forme
18	3HA10	370	370
19	3HA10	282	282
20	3HA10	152	152
21	3HA8	349	349
22	2HA8	349	349
23	19HA6	142	15
24	19HA6	68	51
Barre		Lg/Poids	
HA6		39.9/8.9	
HA8		10.5/4.1	
HA10		24.1/14.9	

Béton

C25/30

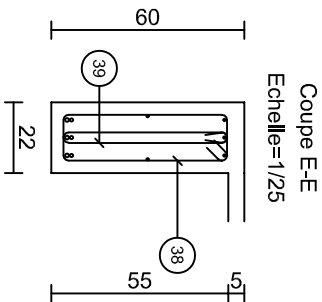
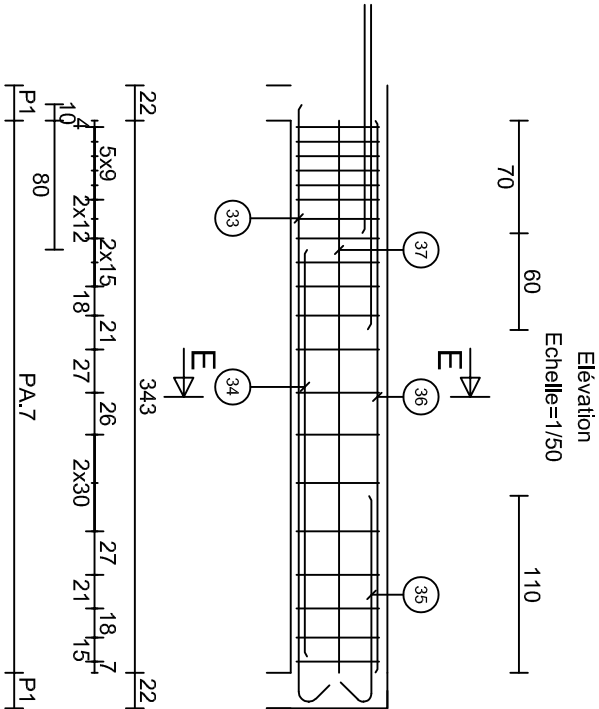
Aciers non soudés

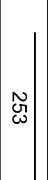
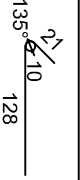
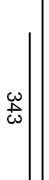
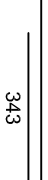
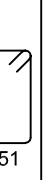
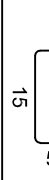
VILLA CHAHIR	PA.7	Béton=0.48 m3 Acier=33.6 kg d=70.7 kg/m3 Ft=8.4 mm Cof=4.7 m² Densité acier: 7850.00 kg / m3	Ed=3.5 cm Eh=5.0 cm Ei=3.5 cm	5
PH RDC				

Section : 22 x 60ht

fck= 25 MPa ftk= 400 MPa Classe de ductilité A Coupe feu R 60

Classe d'exposition: XC1



Barre	Lg	Forme
33	392	
34	253	
35	150	
36	343	
37	343	
38	142	
39	121	

Barre	Lg/Poids
HA6	52.7/11.7
HA8	10.3/4.1
HA10	12.1/7.4
HA12	11.7/10.4

Béton
C25/30

Aciers non
soudés

VILLA CHAHIR
PH RDC

PA.8

Béton=1,07 m3
Acier=104,3 kg d=97,8 kg/m3
Fi=8 mm Coi=10,5 m²
Densité acier: 7850,00 kg / m3

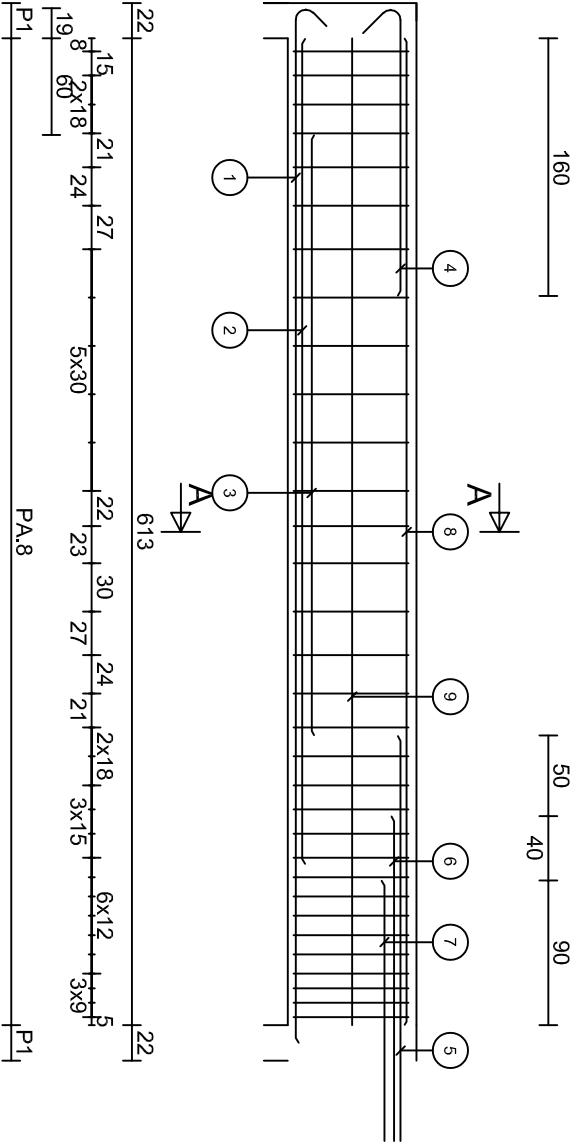
Eb=3,5 cm
Ei=5,0 cm
Ei=3,5 cm

1
4

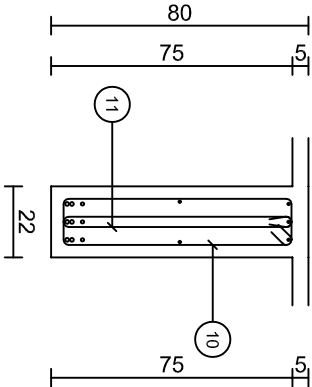
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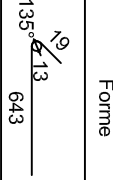
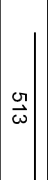
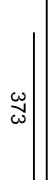
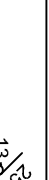
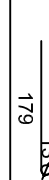
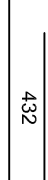
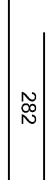
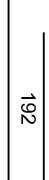
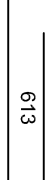
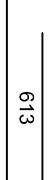
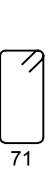
fck= 25 MPa ftk= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1

Elevation
Echelle=1/50



Coupe A-A
Echelle=1/25



Barre	Lg	Forme
1	663	
2	513	
3	373	
4	204	
5	432	
6	282	
7	192	
8	613	
9	613	
10	182	
11	161	
Lg/Poids		
Barre		
HA6	109,8/24,4	
HA8	30,6/12,1	
HA10	11,2/6,9	
HA12	68,6/60,9	

Béton
C25/30

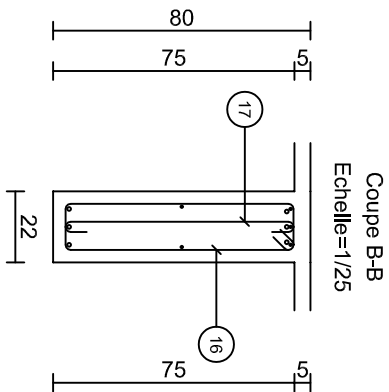
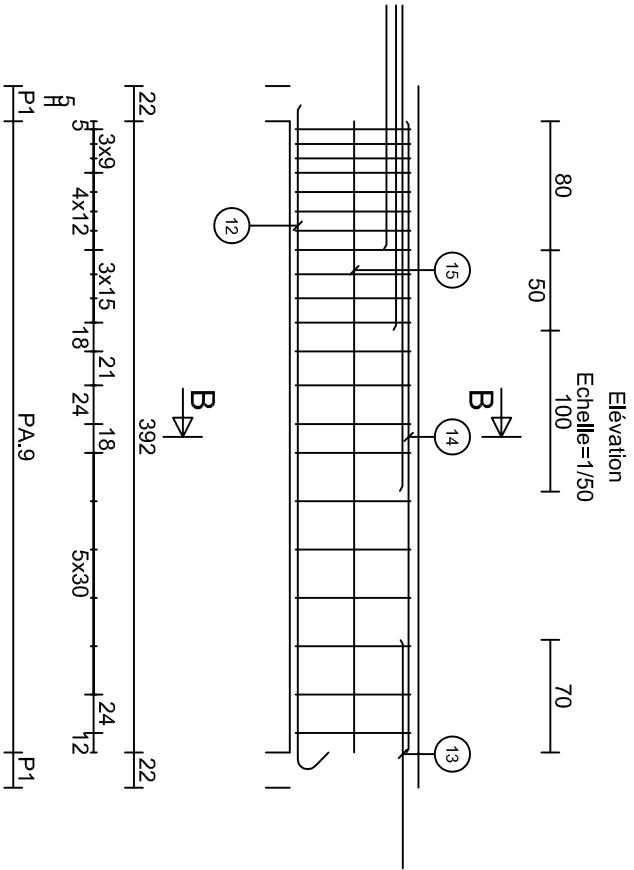
Aciers non
soudés

VILLA CHAHIR	PA.9	Béton=0,68 m3 Acier=34.9 kg d=51.0 kg/m3 F _t =8.9 mm Cof=6.7 m² Densité acier: 7850.00 kg / m3	Ed=3.5 cm Et=5.0 cm Et=3.5 cm	2
PH RDC				4

Section : 22 x 80ht

fck= 25 MPa f_{yk}= 400 MPa Classe de ductilité A Coupe feu R 60

Classe d'exposition: XC1



Barre	Lg	Forme
12	433	413 13.5° 13.5°
13	162	162
14	392	392
15	392	392
16	182	15 12
17	88	71
Barre		Lg/Poids
HA6		56.7/12.6
HA8		19.6/7.7
HA10		4.9/3.0
HA12		13.0/11.5

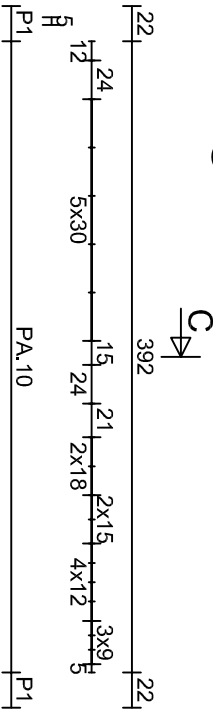
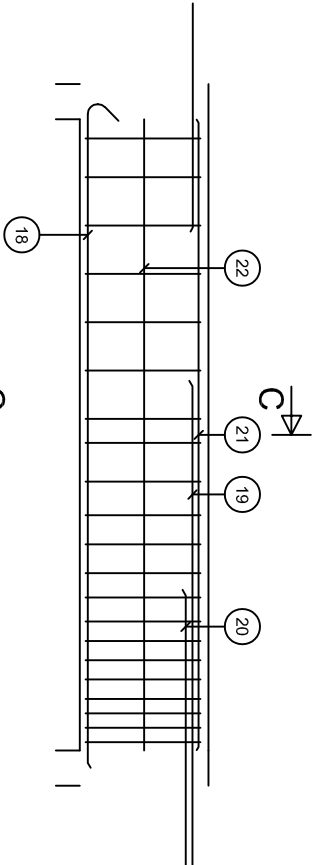
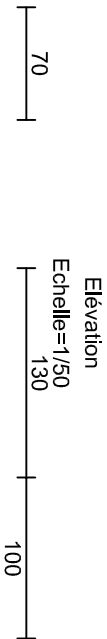
Béton
C25/30

Aciers non
soudés

VILLA CHAHIR	PA.10	Béton=0,68 m3 Acier=55,9 kg d=81,9 kg/m3 Ft=9,2 mm Cof=6,7 m² Densité acier: 7850,00 kg / m3	Ed=3,5 cm Ei=5,0 cm Ei=3,5 cm	3 4
PH RDC				

Section : 22 x 80ht

fck= 25 MPa fyk= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



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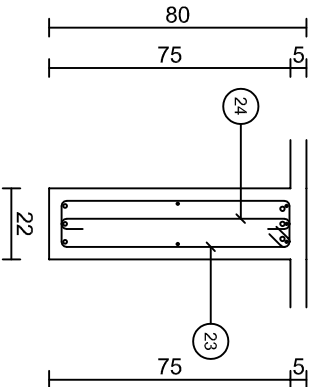
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Barre	Lg	Forme
18	433	
19	432	3HA14 432
20	232	3HA14 232
21	392	3HA8 392
22	392	2HA8 392
23	182	21HA6 182
24	88	21HA6 88

Coupe C-C

Echelle=1/25



Barre	Lg/Poids
HA6	56,7/12,6
HA8	19,6/7,7
HA12	13,0/11,5
HA14	19,9/24,1

Béton
C25/30

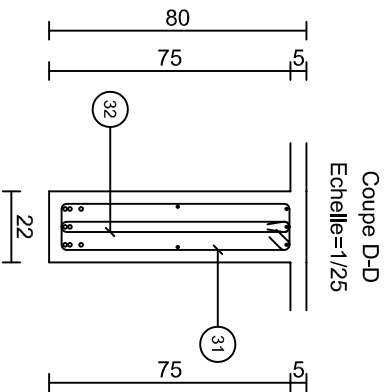
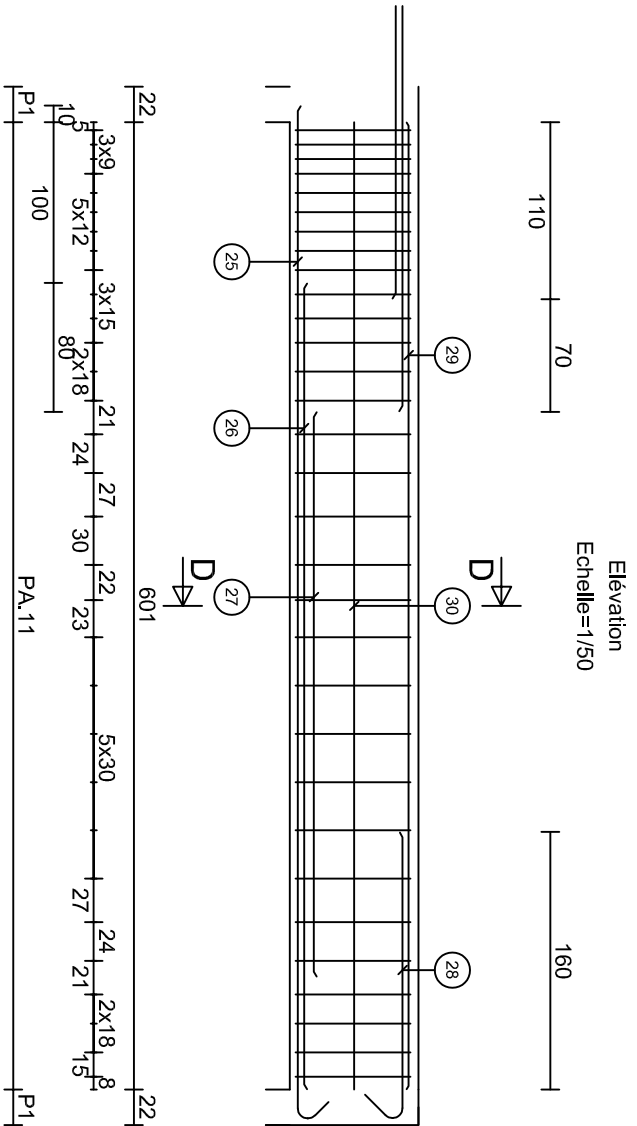
Aciers non
soudés

VILLA CHAHIR	PA.11	Béton=1,05 m3 Acier=77.8 kg d=74.4 kg/m3 Fi=9.0 mm CoF=10.3 m² Densité acier: 7850.00 kg / m3	EB=3.5 cm EI=5.0 cm EI=3.5 cm	4 4
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Section : 22 x 80ht

fck= 25 MPa fyk= 400 MPa Classe de ductilité A Coupe feu R 60

Classe d'exposition: XC1



Barre	Lg	Forme
25 3HA12	650	135° 135° 630 135°
26 3HA12	501	501
27 2HA12	351	351
28 3HA12	204	135° 135° 179 135°
29 3HA8	601	601
30 2HA8	601	601
31 31HA6	182	15 15 15 15
32 31HA6	161	71
Barre		Lg/Poids
HA6		106.4/23.6
HA8		30.0/11.9
HA12		47.7/42.3

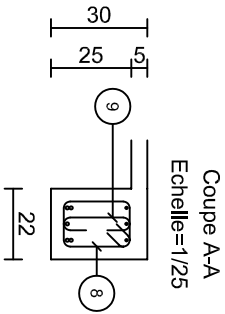
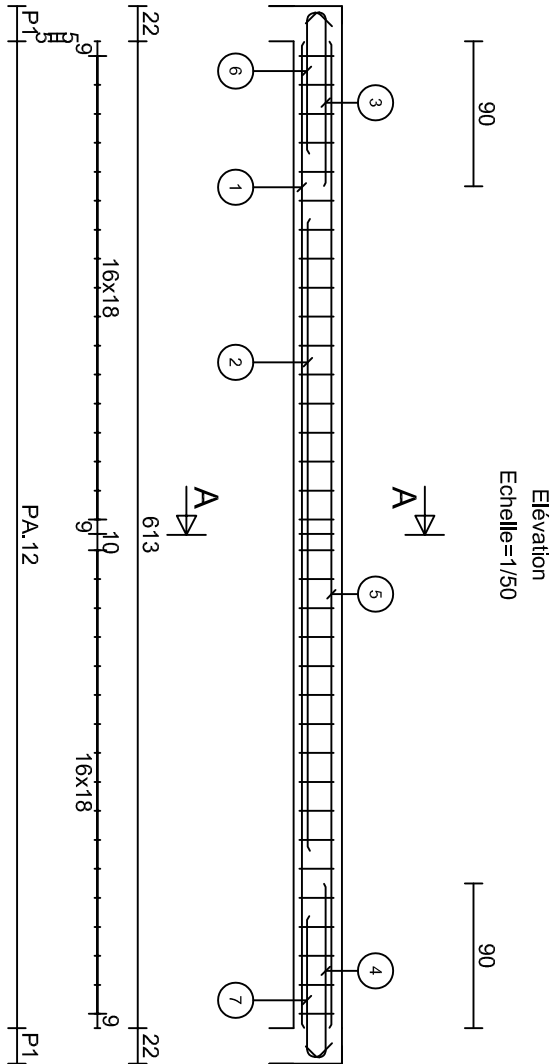
Béton
C25/30

Aciers non
soudés

VILLA CHAHIR PH RDC	PA.12	Béton=0.40 m3 Acier=46.9 kg d=118.0 kg/m3 F _t =8.7 mm Cof=4.7 m² Densité acier: 7850.00 kg / m3	Ed=3.5 cm Ei=5.0 cm Ei=3.5 cm	1 — 1
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Section : 22 x 30ht

fck= 25 MPa f_{yk}= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



Barre	Lg	Forme
1	613	613
2	393	393
3	121	108 8.5 135°
4	121	135° 8 108
5	613	613
6	105	135° 10 88
7	105	10.5 135°
8	82	88 15
9	39	21

Barre	Lg/Poids
HA8	68.0/26.8
HA10	32.5/20.1

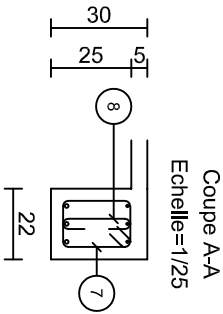
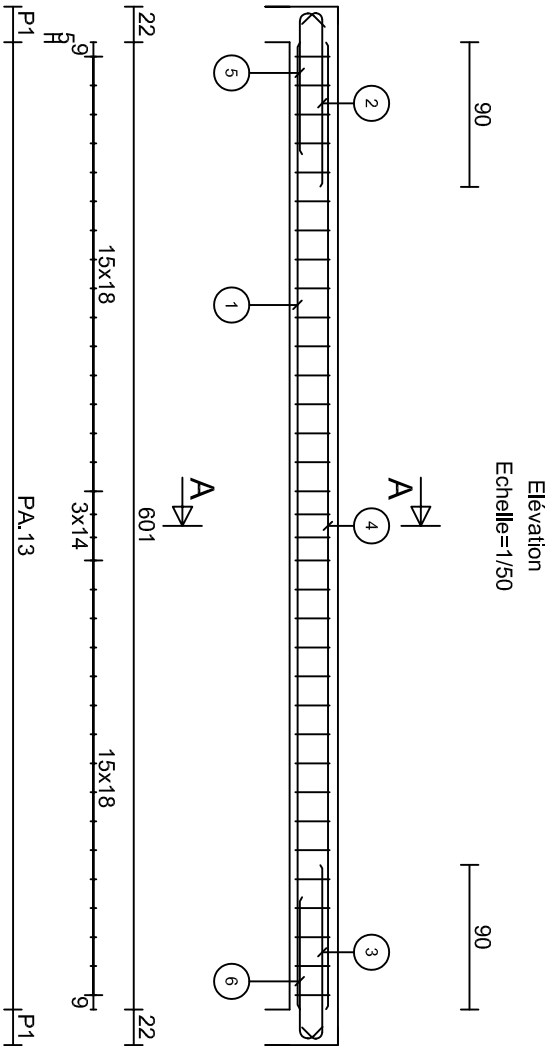
Béton
C25/30

Aciers non
soudés

VILLA CHAHIR	PA.13	Béton=0,39 m3 Acier=38,9 kg d=99,7 kg/m3 F _t =8,4 mm Cof=4,6 m² Densité acier: 7850,00 kg / m3	EB=3,5 cm EI=5,0 cm EI=3,5 cm	1
PH RDC				

Section : 22 x 30ht

fck= 25 MPa f_{yk}= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



Barre	Lg	Forme
1 3HA12	601	601
2 3HA8	121	108 8.5 135°
3 3HA8	121	135° 8.5 108
4 3HA8	601	601
5 3HA10	105	135° 10.5 88
6 3HA10	103	88 10.5 135°
7 34HA6	82	15 21
8 34HA6	38	21

Barre	Lg/Poids
HA6	40.8/9,1
HA8	25.3/10,0
HA10	6.2/3,8
HA12	18.0/16,0

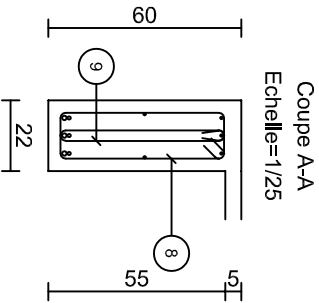
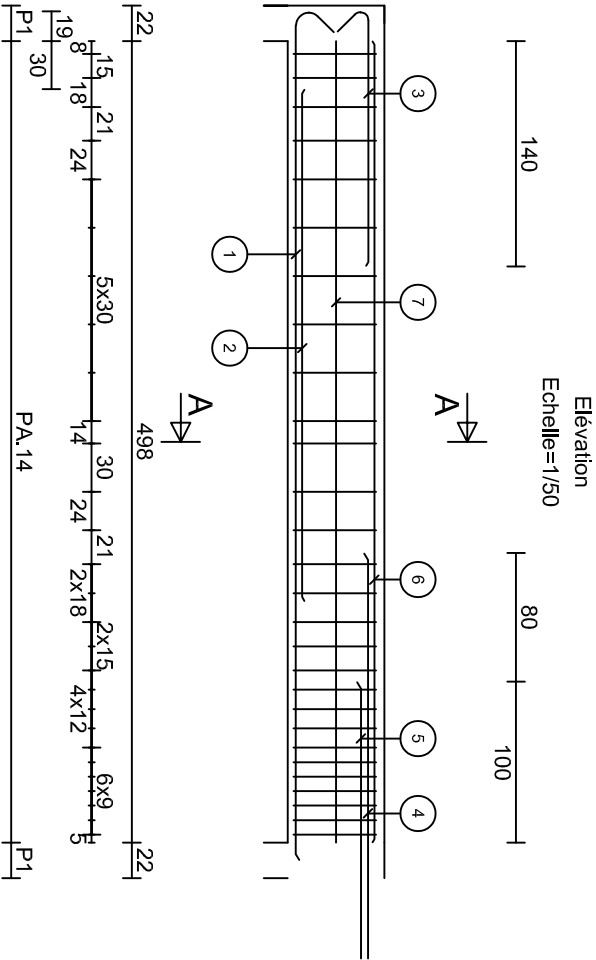
Béton
C25/30

Aciers non
soudés

VILLA CHAHIR	PA.14	Béton=0.67 m3 Acier=80.1 kg d=119.2 kg/m3 Ft=9.4 mm Cof=6.8 m² Densité acier: 7850.00 kg / m3	EB=3.5 cm EI=5.0 cm EI=3.5 cm	1 2
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Section : 22 x 60ht

fck= 25 MPa ftk= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



Barre	Lg	Forme
1 3HA14	551	
2 3HA10	318	
3 3HA10	180	
4 3HA16	382	
5 3HA16	222	
6 3HA8	498	
7 2HA0	498	
8 28HA6	142	
9 28HA6	121	
Barre	Lg/Poids	
HA6 HA8 HA10 HA14 HA16	73.7/16.4 14.9/5.9 14.9/9.2 16.5/20.0 18.1/28.6	

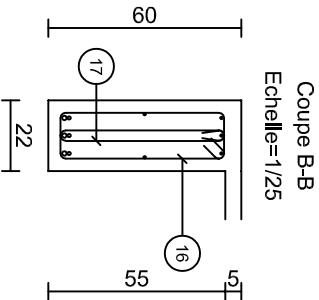
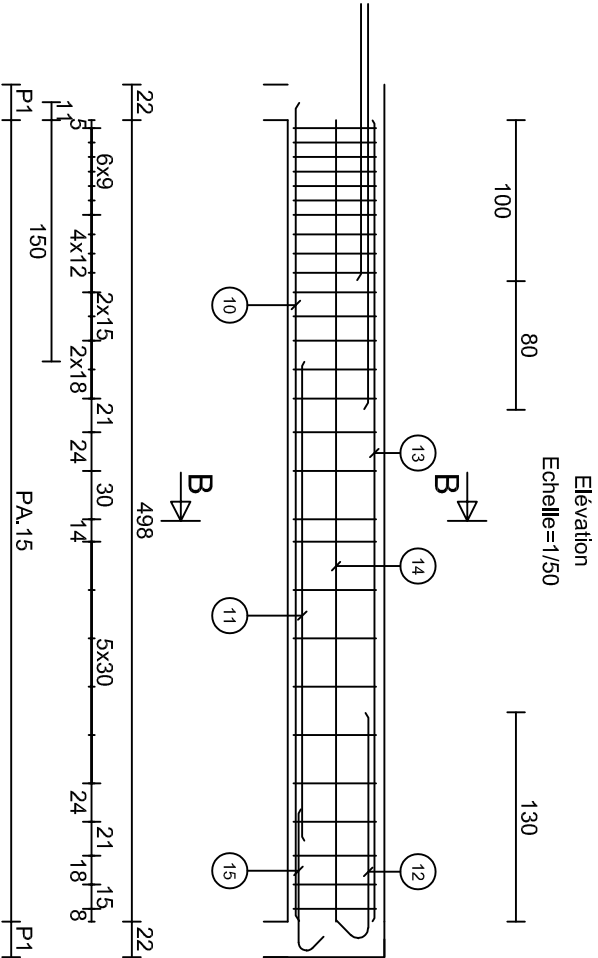
Béton
C25/30

Aciers non
soudés

VILLA CHAHIR	PA.15	Béton=0.67 m3 Acier=52.5 kg d=78.1 kg/m3 Ft=9.4 mm Cof=6.8 m² Densité acier: 7850.00 kg / m3	EB=3.5 cm EI=5.0 cm EI=3.5 cm	2
PH RDC				

Section : 22 x 60ht

fck= 25 MPa ftk= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



Barre	Lg	Forme
10 3HA14	509	509
11 3HA10	298	298
12 3HA12	162	135° 13 141
13 3HA8	498	498
14 2HA0	498	498
15 3HA10	105	105 135° 88
16 28HA6	142	15 15
17 28HA6	121	51
Barre	Lg/Poids	
HA6	73.7/16.4	
HA8	14.9/5.9	
HA10	12.1/7.4	
HA12	4.9/4.3	
HA14	15.3/18.5	

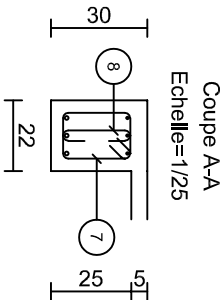
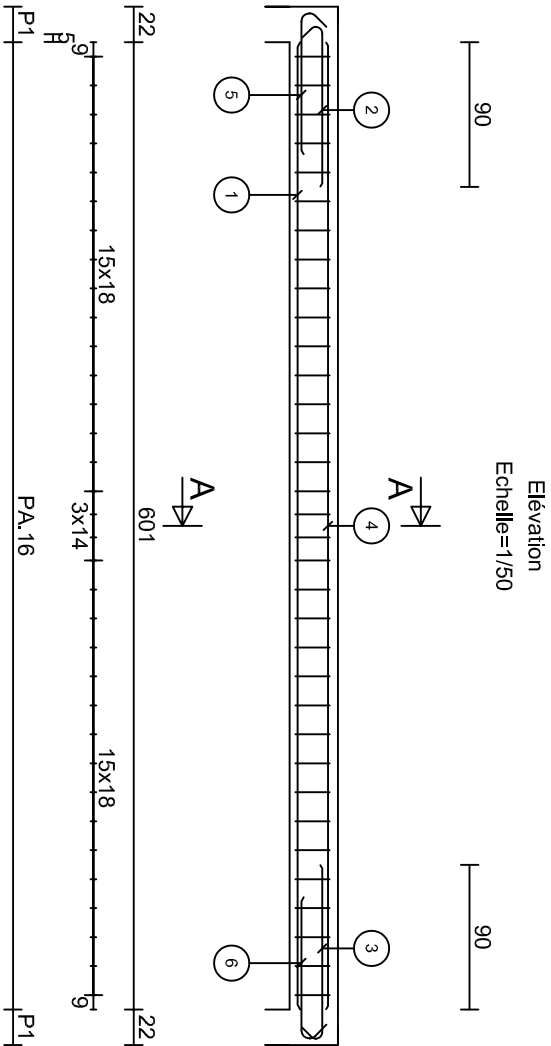
Béton
C25/30

Aciers non
soudés

VILLA CHAHIR PH RDC	PA.16	Béton=0,39 m3 Acier=36,8 kg d=99,3 kg/m3 Ft=8,4 mm Cof=4,6 m² Densité acier: 7850,00 kg / m3	EB=3,5 cm EI=5,0 cm EI=3,5 cm	1 1
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Section : 22 x 30ht

fck= 25 MPa fyk= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



Barre		Lg	Forme
1	3HA12	601	601
2	3HA8	113	100 $\frac{1}{2}$ 8 $\frac{1}{2}$ 135°
3	3HA8	121	135° $\frac{1}{2}$ 8 $\frac{1}{2}$ 135°
4	3HA8	601	601
5	3HA10	105	135° $\frac{1}{2}$ 10 $\frac{1}{2}$ 135°
6	3HA10	105	88 $\frac{1}{2}$ 10 $\frac{1}{2}$ 135°
7	34HA6	82	15 $\frac{1}{2}$ 21
8	34HA6	38	21
Barre		Lg/Poids	
HA6	HA8	40,8/9,1	
HA8	HA10	24,8/9,8	
HA10	HA12	6,3/3,9	
		18,0/16,0	

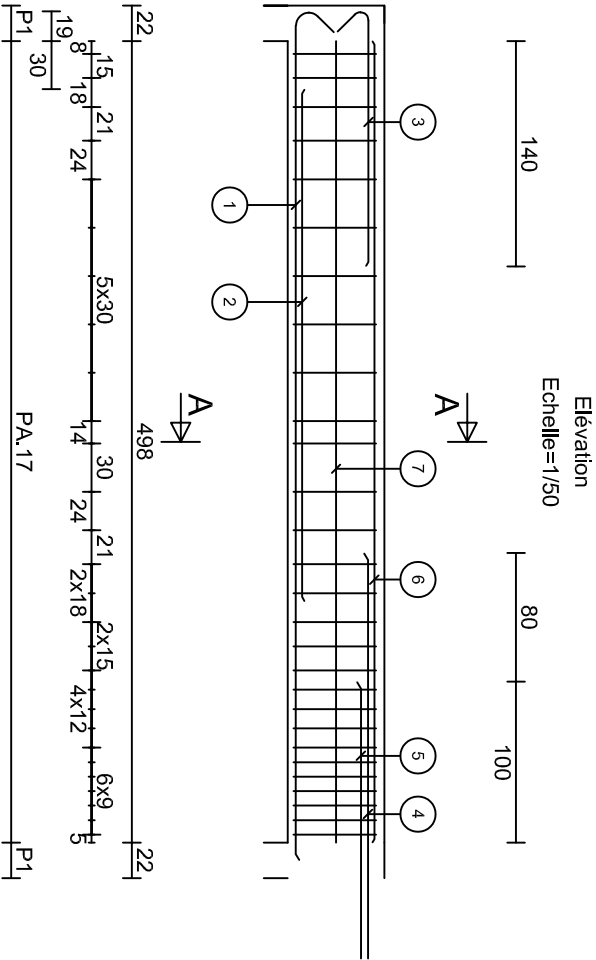
Béton
C25/30

Aciers non
soudés

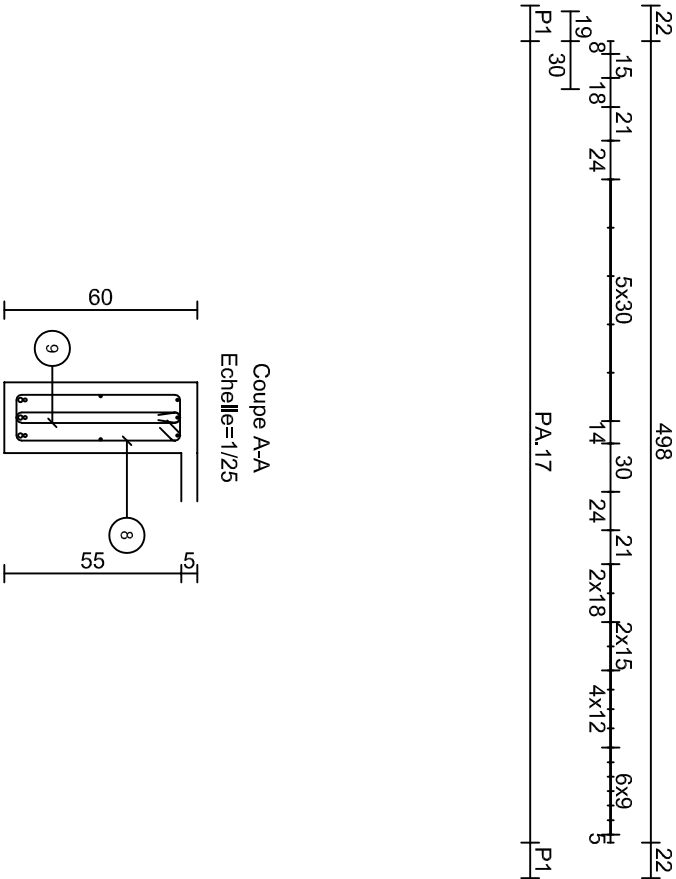
VILLA CHAHIR	PA.17	Béton=0.67 m3 Acier=80.1 kg d=119.2 kg/m3 Ft=9.4 mm Cof=6.8 m² Densité acier: 7850.00 kg / m3	Ed=3.5 cm Et=5.0 cm Et=3.5 cm	1 2
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Section : 22 x 60ht

fck= 25 MPa ftk= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



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*
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Barre	Lg	Forme
1 3HA14	551	
2 3HA10	318	
3 3HA10	180	
4 3HA16	382	
5 3HA16	222	
6 3HA8	498	
7 2HA0	498	
8 28HA6	142	
9 28HA6	121	
Barre	Lg/Poids	
HA6 HA8 HA10 HA14 HA16	73.7/16.4 14.9/5.9 14.9/9.2 16.5/20.0 18.1/28.6	

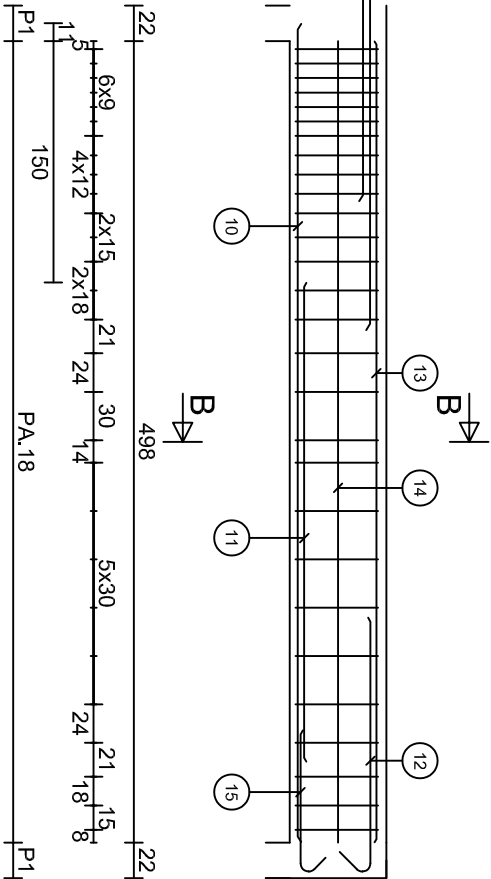
Béton
C25/30

Aciers non
soudés

VILLA CHAHIR	PA.18	Béton=0,67 m3 Acier=51,5 kg d=76,6 kg/m3 Ft=9,3 mm Cof=6,8 m² Densité acier: 7850,00 kg / m3	EB=3,5 cm EI=5,0 cm EI=3,5 cm	2
PH RDC				

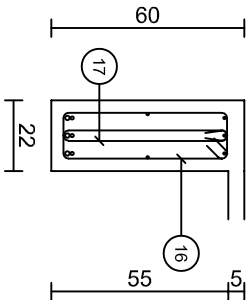
Section : 22 x 60ht

fck= 25 MPa ftk= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



Coupe B-B

Echelle=1/25



Barre	Lg	Forme
10	509	509
11	298	298
12	180	135° 10/10 158
13	498	498
14	498	498
15	105	88 10/5 135°
16	142	15 15
17	121	51
Barre	Lg/Poids	
HA6	73,7/16,4	
HA8	14,9/5,9	
HA10	17,5/10,8	
HA14	15,3/18,5	

Béton

C25/30

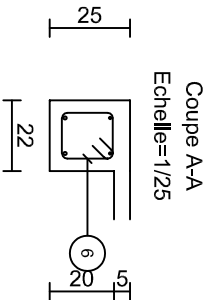
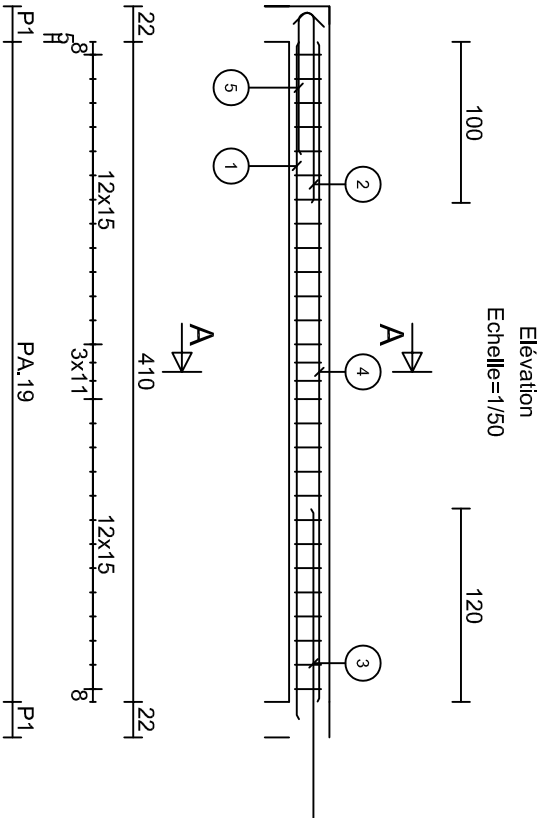
Aciers non soudés

Poutre n09 Niveau n01	PA.19	Béton=0,22 m3 Acier=18.5 kg d=84.3 kg/m3 Ft=8.0 mm Cof=2.7 m² Densité acier: 7850.00 kg / m3	Eb=3.5 cm Ei=5.0 cm Ei=3.5 cm	1 2
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Section : 22 x 25ht

fck= 25 MPa fyk= 400 MPa Classe de ductilité A Coupe feu R 60

Classe d'exposition: XC1



Barre		Lg	Forme
1	2HA10	421	421
2	2HA8	131	118 85 135°
3	2HA10	262	262
4	2HA8	410	410
5	2HA10	105	135° 10 88
6	28HA6	72	15

Barre		Lg/Poids
HA6	HA8	20.3/4.5
HA8	HA10	10.8/4.3
HA10		15.8/9.7

Béton

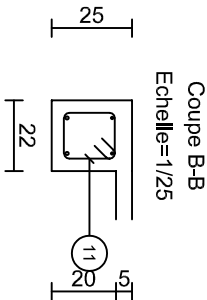
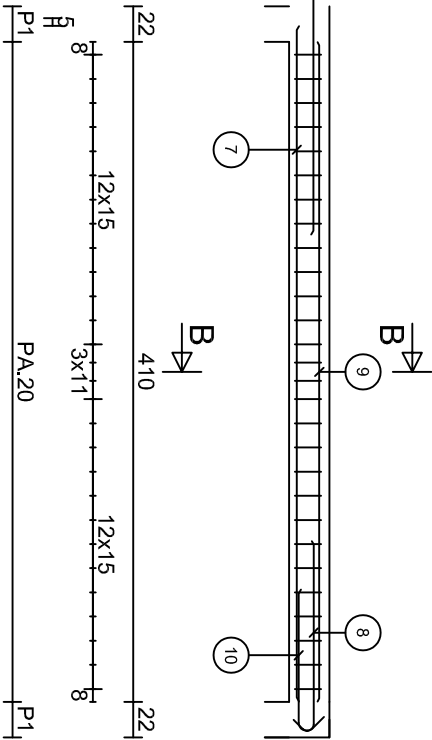
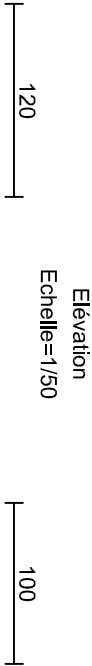
C25/30

Aciers non soudés

Poutre n09 Niveau n01	PA.20	Béton=0,22 m3 Acier=15,2 kg d=69,5 kg/m3 Ft=8,0 mm Cof=2,7 m² Densité acier: 7850,00 kg / m3	EB=3,5 cm EI=5,0 cm EI=3,5 cm	2
PH RDC				2

Section : 22 x 25ht

fck= 25 MPa fyk= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



Barre		Lg	Forme
7	2HA10	420	420
8	2HA8	131	135° 118 135°
9	2HA8	410	410
10	2HA10	105	105° 88 135°
11	28HA6	72	15 15
Barre		Lg/Poids	
HA6		20.3/4.5	
HA8		10.8/4.3	
HA10		10.5/6.5	

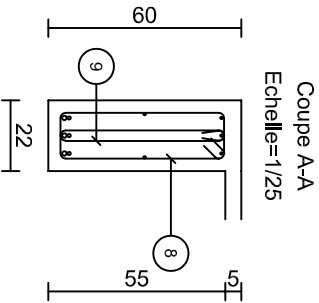
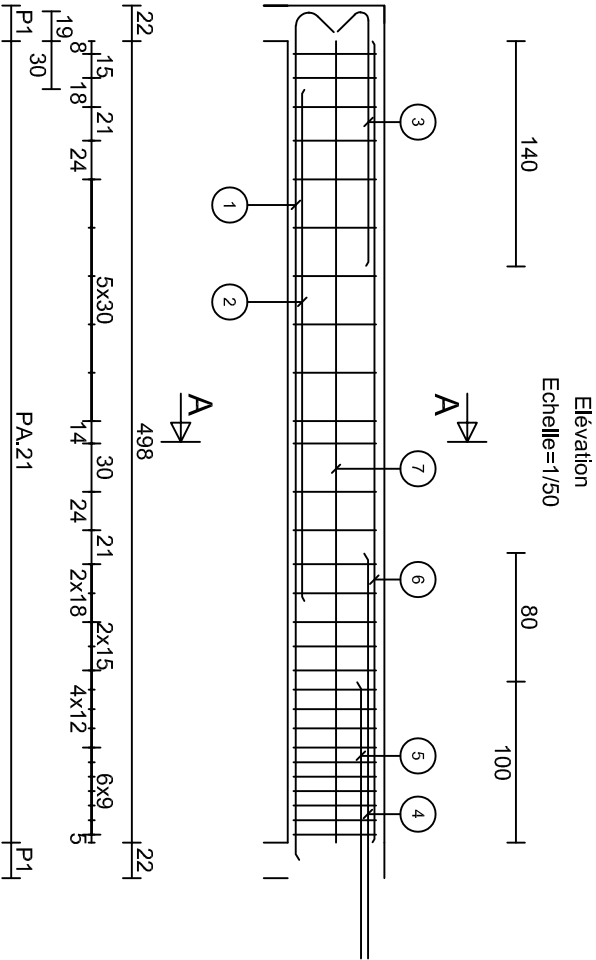
Béton
C25/30

Aciers non
soudés

VILLA CHAHIR	PA.21	Béton=0.67 m3 Acier=80.1 kg d=119.2 kg/m3 Ft=9.4 mm Cof=6.8 m² Densité acier: 7850.00 kg / m3	Eb=3.5 cm Et=5.0 cm Et=3.5 cm	1 2
PH RDC				

Section : 22 x 60ht

fck= 25 MPa ftk= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



Barre	Lg	Forme
1 3HA14	551	
2 3HA10	318	
3 3HA10	180	
4 3HA16	382	
5 3HA16	222	
6 3HA8	498	
7 2HA0	498	
8 28HA6	142	
9 28HA6	121	
Barre	Lg/Poids	
HA6	73.7/16.4	
HA8	14.9/5.9	
HA10	14.9/9.2	
HA14	16.5/20.0	
HA16	18.1/28.6	

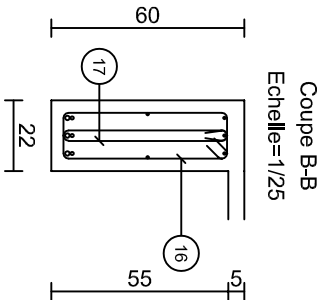
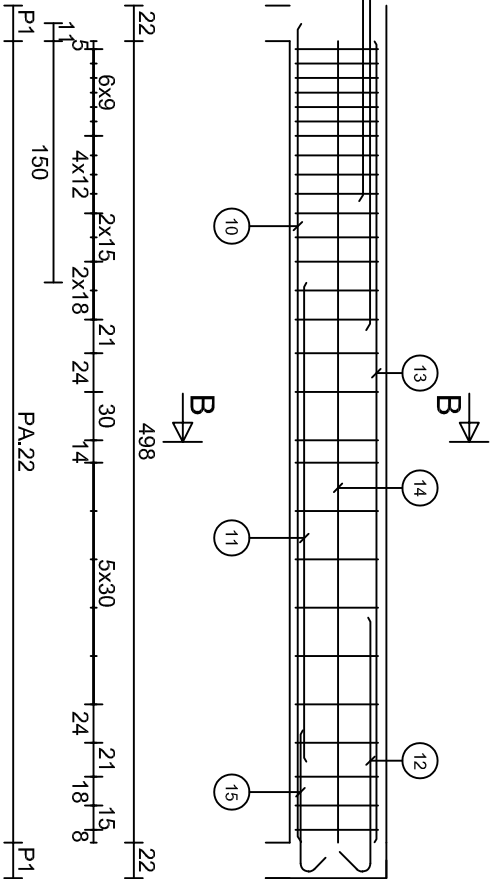
Béton
C25/30

Aciers non
soudés

VILLA CHAHIR	PA.22	Béton=0.67 m3 Acier=51.5 kg d=76.6 kg/m3 Ft=9.3 mm Cof=6.8 m² Densité acier: 7850.00 kg / m3	EB=3.5 cm EI=5.0 cm EI=3.5 cm	2
PH RDC				

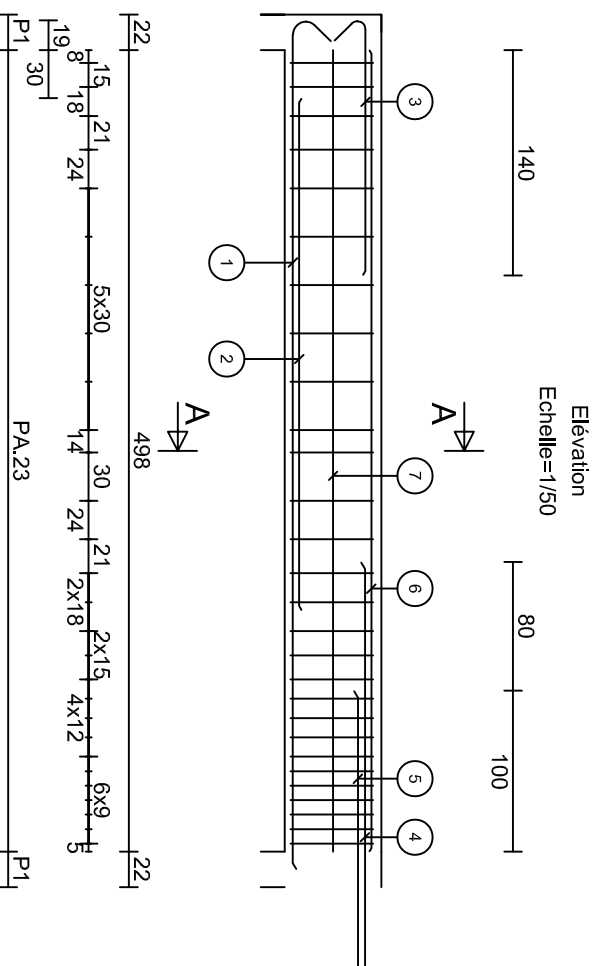
Section : 22 x 60ht

fck= 25 MPa ftk= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1

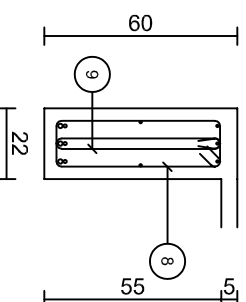


Barre	Lg	Forme
10	509	509
11	298	298
12	180	135° 10 158
13	498	498
14	498	498
15	105	88 10 135°
16	142	15 15
17	121	51
Barre	Lg/Poids	
HA6	73.7/16.4	
HA8	14.9/5.9	
HA10	17.5/10.8	
HA14	15.3/18.5	

Section : 22 x 60ht	
fck= 25 MPa	fyk= 400 MPa
Classe de ductilité A	Coupe feu R 60
Classe d'exposition: XC1	



Coupe A-A
Echelle=1/25



	Barre	Lg	Forme
1	3HA14	551	
2	3HA10	318	
3	3HA10	180	
4	3HA16	382	
5	3HA16	222	
6	3HA8	498	
7	2HA0	498	
8	28HA6	142	
9	28HA6	121	

Béton
C25/30

Aciers non
soudés

VILLA CHAHIR
PH RDC

PA.24

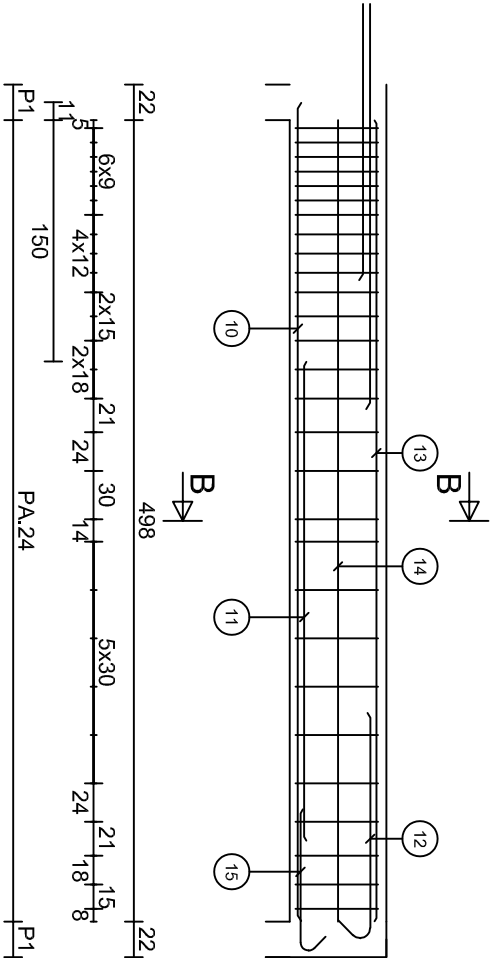
Béton=0.67 m3
Acier=52.5 kg d=78.1 kg/m3
Ft=9.4 mm Cof=6.8 m²
Densité acier: 7850.00 kg / m3

EB=3.5 cm
EI=5.0 cm
EI=3.5 cm

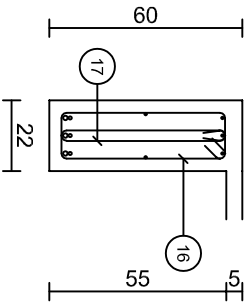
2

Section : 22 x 60ht

fck= 25 MPa ftk= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



Coupe B-B
Echelle=1/25



Barre	Lg	Forme
10	509	509
11	298	298
12	162	135° 135° 141
13	498	498
14	498	498
15	105	88 105 135°
16	142	15 15
17	121	51

Barre	Lg/Poids
HA6	73.7/16.4
HA8	14.9/5.9
HA10	12.1/7.4
HA12	4.9/4.3
HA14	15.3/18.5

Béton
C25/30

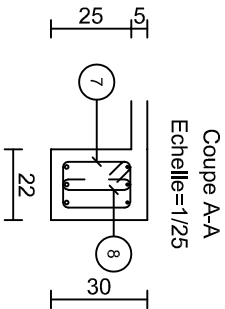
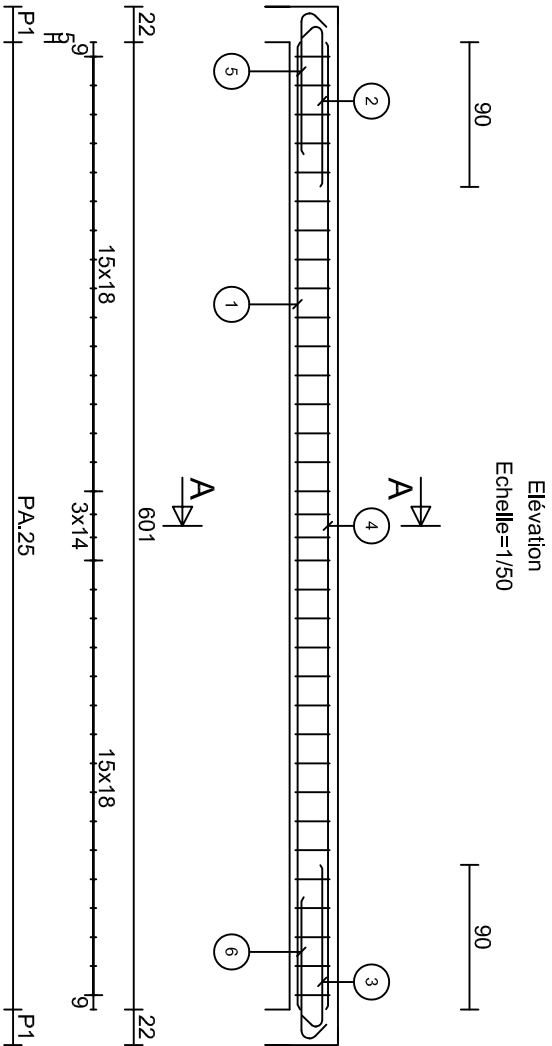
Aciers non
soudés

VILLA CHAHIR PH RDC	PA.25	Béton=0,39 m3 Acier=36,8 kg d=99,3 kg/m3 F _t =8,4 mm Cof=4,6 m² Densité acier: 7850,00 kg / m3	EB=3,5 cm EI=5,0 cm EI=3,5 cm	1 1
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Section : 22 x 30ht

fck= 25 MPa f_{yk}= 400 MPa Classe de ductilité A Coupe feu R 60

Classe d'exposition: XC1



Barre		Lg	Forme
1	3HA12	601	601
2	3HA8	113	100 85 135°
3	3HA8	114	135° 85 101
4	3HA8	601	601
5	3HA10	105	135° 105 88
6	3HA10	105	88 105 135°
7	34HA6	82	15 21
8	34HA6	38	21

Barre		Lg/Poids
HA6	HA8	40,8/9,1
HA8	HA10	24,8/9,8
HA10	HA12	6,3/3,9
HA12		18,0/16,0

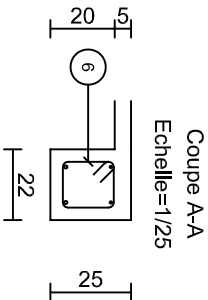
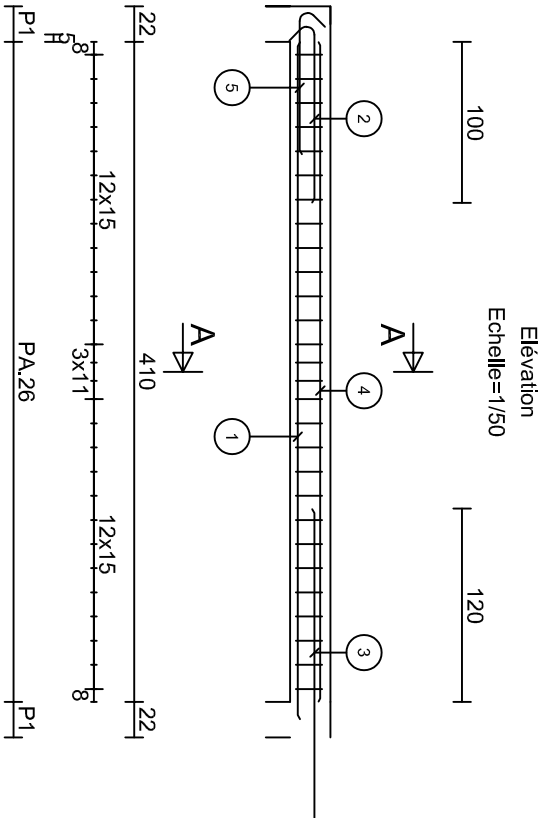
Béton
C25/30

Aciers non
soudés

VILLA CHAHIR PH RDC	PA.26	Béton=0,22 m3 Acier=19.0 kg d=86.7 kg/m3 Ft=8.1 mm Cof=2.7 m² Densité acier: 7850.00 kg / m3	Ed=3.5 cm Et=5.0 cm Et=3.5 cm	1 2
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Section : 22 x 25ht

fck= 25 MPa ftk= 400 MPa Classe de ductilité A Coupe feu R 60 | Classe d'exposition: XC1



Barre		Lg	Forme
1	2HA10	421	421
2	2HA10	126	110 102 135°
3	2HA10	262	262
4	2HA8	410	410
5	2HA10	105	135° 102 88
6	28HA6	72	15 9°

Barre		Lg/Poids
HA6	HA8	20.3/4.5
HA8	HA10	8.2/3.2
HA10		18.3/11.3

Béton

C25/30

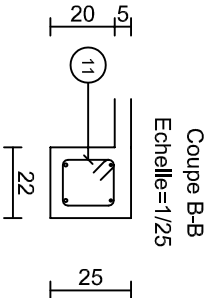
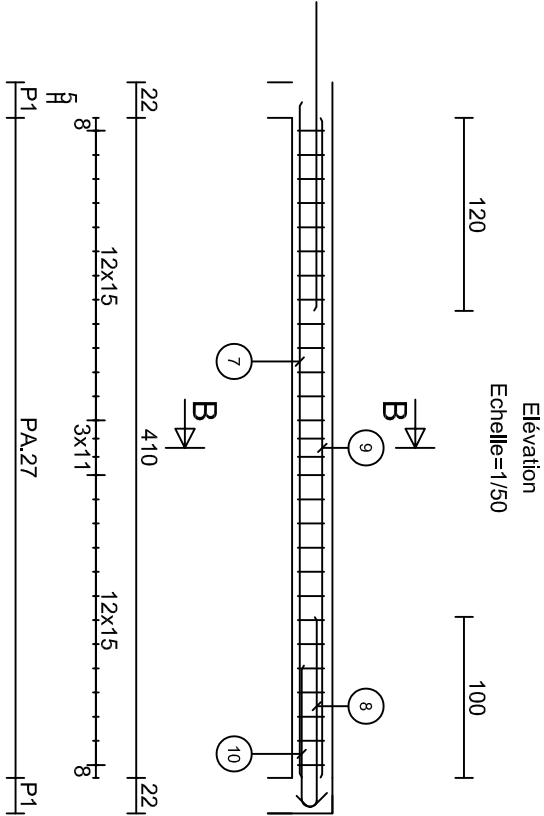
Aciers non soudés

VILLA CHAHIR	PA.27	Béton=0,22 m3	EB=3,5 cm	2
PH RDC		Acier=15,2 kg d=69,5 kg/m3	EI=5,0 cm	
		Ft=8,0 mm Cof=2,7 m²	EI=3,5 cm	2
		Densité acier: 7850,00 kg / m3		

Section : 22 x 25ht

fck= 25 MPa ftk= 400 MPa Classe de ductilité A Coupe feu R 60

Classe d'exposition: XC1



Barre		Lg	Forme
7	2HA10	420	420
8	2HA8	131	$\frac{N}{8}$ 135° 118
9	2HA8	410	410
10	2HA10	105	$\frac{N}{5}$ 105 135°
11	28HA6	72	15
Barre		Lg/Poids	
HA6		20,3/4,5	
HA8		10,8/4,3	
HA10		10,5/6,5	